Rounding and Estimating

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS round *(whakaawhiwhi)* · decimal place *(mati ā-ira)* · significant figure *(mati hira)*

A Rounding whole numbers ●○○

Remember: Look at the digit to the right of the place you are rounding to. If it is 5 or more, round up. If it is 4 or less, round down.

1. Round each number.

a) 47 (nearest 10) _____ **b)** 382 (nearest 10) _____ **c)** 382 (nearest 100) _____

d) 1,650 (nearest 100) _____ **e)** 7,499 (nearest 1,000) _____ **f)** 12,850 (nearest 1,000) _____

g) 995 (nearest 10) _____ **h)** 64,500 (nearest 1,000) _____ **i)** 2,049 (nearest 100) _____

j) 150 (nearest 100) _____ **k)** 9,960 (nearest 100) _____ **l)** 449,999 (nearest 1,000) _____

B Decimal places ●○○

Remember: Count the decimal places after the decimal point. 3.14159 to 2 decimal places (2 d.p.) is 3.14.

2. Round each number to the number of decimal places shown.

a) 4.37 (1 d.p.) _____ **b)** 12.84 (1 d.p.) _____ **c)** 0.652 (2 d.p.) _____

d) 7.918 (2 d.p.) _____ **e)** 3.5 (nearest whole number) **f)** 26.449 (1 d.p.) _____

g) 0.0862 (3 d.p.) _____ **h)** 9.96 (1 d.p.) _____ **i)** 15.995 (2 d.p.) _____

j) 0.4444 (2 d.p.) _____ **k)** 81.25 (1 d.p.) _____ **l)** 2.7182 (3 d.p.) _____

C Significant figures ●●○

Remember: Count significant figures from the first digit that is not zero. In 0.00452 the first significant figure is the 4.

3. How many significant figures does each number have?

a) 47 _____ **b)** 3.08 _____ **c)** 0.0052 _____ **d)** 12.40 _____

e) 600.5 _____ **f)** 0.070 _____ **g)** 9 _____ **h)** 1.005 _____

4. Round each number to the number of significant figures shown.

a) 4,827 (2 s.f.) _____ **b)** 36.48 (3 s.f.) _____ **c)** 0.07251 (2 s.f.) _____

d) 592 (1 s.f.) _____ **e)** 18.049 (3 s.f.) _____ **f)** 0.00386 (1 s.f.) _____

g) 27,450 (2 s.f.) _____ **h)** 6.995 (3 s.f.) _____ **i)** 0.5078 (2 s.f.) _____

j) 8,049 (1 s.f.) _____ **k)** 0.09961 (2 s.f.) _____ **l)** 3,001.8 (3 s.f.) _____

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D Estimating ●●●

> **Remember:** Round each number to 1 significant figure, then calculate. The answer is close to the exact answer and quick to find.

5. Estimate the answer to each calculation by rounding each number to 1 significant figure.

a) 38 × 21

b) 612 × 49

c) 7.8 × 5.2

d) 283 + 619

e) 792 ÷ 38

f) 58.6 ÷ 1.9

g) 4,120 × 0.48

h) 8,950 ÷ 31

6. Estimate the answer to each problem by rounding each number to 1 significant figure.

a) A hall has 38 rows of 21 seats. About how many seats is that?

b) One book costs $18.95. About how much do 29 books cost?

c) A car travels 412 km on 38 litres of petrol. About how many kilometres is that for each litre?

d) 8,920 people share 31 buses equally. About how many people are on each bus?

★ Extension

7. A length is 8 cm, to the nearest centimetre. Write the smallest and largest values the length could be.

8. A bag has a mass of 4.5 kg, to 1 decimal place. Write the smallest and largest values the mass could be.

9. A crowd is 350 people, to the nearest 10. Write the smallest and largest values the crowd could be.

10. A distance is 1200 m, to the nearest 100 m. Write the smallest and largest values the distance could be.

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Rounding and Estimating

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS round *(whakaawhiwhi)* · decimal place *(mati ā-ira)* · significant figure *(mati hira)*

A Rounding whole numbers ●○○

Remember: Look at the digit to the right of the place you are rounding to. If it is 5 or more, round up. If it is 4 or less, round down.

1. Round each number.

a) 63 (nearest 10) _____ **b)** 527 (nearest 10) _____ **c)** 527 (nearest 100) _____

d) 2,350 (nearest 100) _____ **e)** 8,501 (nearest 1,000) _____ **f)** 23,490 (nearest 1,000) _____

g) 896 (nearest 10) _____ **h)** 37,500 (nearest 1,000) _____ **i)** 4,951 (nearest 100) _____

j) 250 (nearest 100) _____ **k)** 8,970 (nearest 100) _____ **l)** 549,999 (nearest 1,000) _____

B Decimal places ●○○

Remember: Count the decimal places after the decimal point. 3.14159 to 2 decimal places (2 d.p.) is 3.14.

2. Round each number to the number of decimal places shown.

a) 6.23 (1 d.p.) _____ **b)** 15.78 (1 d.p.) _____ **c)** 0.847 (2 d.p.) _____

d) 3.926 (2 d.p.) _____ **e)** 8.5 (nearest whole number) **f)** 41.349 (1 d.p.) _____

g) 0.0537 (3 d.p.) _____ **h)** 7.97 (1 d.p.) _____ **i)** 24.996 (2 d.p.) _____

j) 0.7777 (2 d.p.) _____ **k)** 63.75 (1 d.p.) _____ **l)** 1.4142 (3 d.p.) _____

C Significant figures ●●○

Remember: Count significant figures from the first digit that is not zero. In 0.00452 the first significant figure is the 4.

3. How many significant figures does each number have?

a) 63 _____ **b)** 5.02 _____ **c)** 0.0037 _____ **d)** 18.60 _____

e) 400.2 _____ **f)** 0.090 _____ **g)** 4 _____ **h)** 2.008 _____

4. Round each number to the number of significant figures shown.

a) 6,283 (2 s.f.) _____ **b)** 52.37 (3 s.f.) _____ **c)** 0.04672 (2 s.f.) _____

d) 748 (1 s.f.) _____ **e)** 27.058 (3 s.f.) _____ **f)** 0.00529 (1 s.f.) _____

g) 43,650 (2 s.f.) _____ **h)** 8.996 (3 s.f.) _____ **i)** 0.3096 (2 s.f.) _____

j) 6,051 (1 s.f.) _____ **k)** 0.08962 (2 s.f.) _____ **l)** 5,002.7 (3 s.f.) _____

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D Estimating ●●●

> **Remember:** Round each number to 1 significant figure, then calculate. The answer is close to the exact answer and quick to find.

5. Estimate the answer to each calculation by rounding each number to 1 significant figure.

a) 42 × 19

b) 587 × 31

c) 6.9 × 4.1

d) 372 + 518

e) 612 ÷ 29

f) 79.4 ÷ 3.8

g) 5,870 × 0.52

h) 7,940 ÷ 42

6. Estimate the answer to each problem by rounding each number to 1 significant figure.

a) A hall has 42 rows of 19 seats. About how many seats is that?

b) One book costs $21.95. About how much do 38 books cost?

c) A car travels 587 km on 32 litres of petrol. About how many kilometres is that for each litre?

d) 7,940 people share 42 buses equally. About how many people are on each bus?

★ Extension

7. A length is 6 cm, to the nearest centimetre. Write the smallest and largest values the length could be.

8. A bag has a mass of 7.2 kg, to 1 decimal place. Write the smallest and largest values the mass could be.

9. A crowd is 480 people, to the nearest 10. Write the smallest and largest values the crowd could be.

10. A distance is 2500 m, to the nearest 100 m. Write the smallest and largest values the distance could be.

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